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What Cannot Be Known, Because It Does Not Exist

Uncertainty and Complementarity

A police officer pulls over Werner Heisenberg for speeding. “Do you know how fast you were going?” asks the cop. “No,” Heisenberg replies, “but I know exactly where I am!”

I think we can all agree that physics jokes are the funniest jokes there are. They are less good at accurately conveying physics. This particular chestnut rests on familiarity with the famous Heisenberg uncertainty principle, often explained as saying that we cannot simultaneously know both the position and the velocity of any object. But the reality is deeper than that.

It’s not that we can’t know position and momentum, it’s that they don’t even exist at the same time. Only under extremely special circumstances can an object be said to have a location—when its wave function is entirely concentrated on one point in space, and zero everywhere else—and similarly for velocity. And when one of the two is precisely defined, the other could be literally anything, were we to measure it. More often, the wave function includes a spread of possibilities for both quantities, so neither has a definite value.

Back in the 1920s, all this was less clear. It was still natural to think that the probabilistic nature of quantum mechanics simply indicated that it was an incomplete theory, and that there was a more deterministic, classical-sounding picture waiting to be developed. Wave functions, in other words,